3▶

$$\int u' \sec^2 u \, dx = \tan u + c$$

$$\int u' \operatorname{cosec}^2 u \, dx = -\cot u + c$$

$$\int u' \tan^2 u \, dx = \int u' (\sec^2 u - 1) \, dx = \tan u - u + c$$

$$\int u' \cot^2 u \, dx = \int u' (\operatorname{cosec}^2 u - 1) \, dx = -\cot u - u + c$$

$$\int u' \sin^2 u \, dx = \frac{1}{2}\int u' (1 - \cos 2u) \, dx = \frac{1}{2}\left(u - \frac{\sin 2u}{2}\right) + c$$

$$\int u' \cos^2 u \, dx = \frac{1}{2}\int u' (1 + \cos 2u) \, dx = \frac{1}{2}\left(u + \frac{\sin 2u}{2}\right) + c$$

4▶

$$\int \frac{u'}{a^2 + u^2} \, dx = \frac{1}{a} \tan^{-1}\left(\frac{u}{a}\right) + c$$

$$\int \frac{u'}{\sqrt{a^2 - u^2}} \, dx = \sin^{-1}\left(\frac{u}{a}\right) + c$$

$$\int \frac{u'}{u\sqrt{u^2 - a^2}} \, dx = \frac{1}{a} \sec^{-1}\left(\frac{u}{a}\right) + c$$

$$\int \frac{u'}{a^2 - u^2} \, dx = \frac{1}{a} \tanh^{-1}\left(\frac{u}{a}\right) + c$$

$$\int \frac{u'}{\sqrt{a^2 + u^2}} \, dx = \sinh^{-1}\left(\frac{u}{a}\right) + c$$

$$\int \frac{u'}{\sqrt{u^2 - a^2}} \, dx = \cosh^{-1}\left(\frac{u}{a}\right) + c$$

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Lec. Problems

(14) $I = \int \tan^3 7x \sec^2 7x \, dx$

$= \int \sec^2 7x \, (\tan 7x)^3 \, dx$

$= \frac{1}{7} \int 7 \sec^2 7x \, (\tan 7x)^3 \, dx$

$= \frac{1}{7} \frac{(\tan 7x)^4}{4} + C$

(23) $I = \int \frac{1}{x + x \ln x} \, dx$

$= \int \frac{1}{x(1 + \ln x)} \, dx$

$= \int \frac{1/x}{(1 + \ln x)} \, dx$ $\left(\int \frac{u'}{u} dx = \ln|u| + C \right)$

$= \ln |1 + \ln x| + C$

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(6) $I = \int \sec^2 3x \, dx$

$= \frac{1}{3} \int 3 \sec^2 (3x) \, dx$

$= \frac{1}{3} \tan 3x + C$

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(20) $I = \int (3 + 4\cos x)^2 \, dx$

$= \int 9 + 24\cos x + 16 \cos^2 x \, dx$

$= \int 9 + 24\cos x + 16 \times \frac{1}{2}(1 + \cos 2x) \, dx$

$= 9x + 24\sin x + 8\left(x + \frac{\sin 2x}{2}\right) + C$

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(21) $I = \int \frac{1+\sin x}{\cos^2 x}\, dx$

$= \int \sec^2 x + \sin x (\cos x)^{-2}\, dx$

$= \int \sec^2 x - (-\sin x)(\cos x)^{-2}\, dx$

$= \tan x - \frac{(\cos x)^{-1}}{-1} + C$

(22) $I = \int \frac{1-\cos 2x}{\sin^2 x}\, dx$

$= \int \frac{\cancel{1-\cos 2x}}{\frac{1}{2}(\cancel{1-\cos 2x})}\, dx$

$= \int 2\, dx = 2x + C$

(*) $I = \int \frac{x}{x^2+9}\, dx$

$= \frac{1}{2}\int \frac{2x}{x^2+9}\, dx$

$= \frac{1}{2} \ln|x^2+9| + C$

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(*) $I = \int 3 \sin^2 x \, dx$

$= 3 \int \frac{1}{2}(1 - \cos 2x) \, dx$

$= \frac{3}{2} \left[x - \frac{\sin 2x}{2} \right] + C$

(*) $I = \int 5 \cos^2 4x \, dx$

$= 5 \int \frac{1}{2}(1 + \cos 8x) \, dx$

$= \frac{5}{2} \left[x + \frac{\sin 8x}{8} \right] + C$

(*) $I = \int 3 \tan^2 6x \, dx$

$= 3 \int \sec^2 6x - 1 \, dx$

$= 3 \left[\frac{\tan 6x}{6} - x \right] + C$

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(24) $I = \int \frac{1}{x^2+9}\,dx$

$= \int \frac{1}{x^2+3^2}\,dx$ $\qquad \int \frac{u'}{u^2+a^2}\,dx = \frac{1}{a}\tan^{-1}\left(\frac{u}{a}\right)+C$

$= \frac{1}{3}\tan^{-1}\left(\frac{x}{3}\right)+C$

(28) $I = \int \frac{\cos x}{\sqrt{4-\sin^2 x}}\,dx$

$= \int \frac{\cos x}{\sqrt{2^2-(\sin x)^2}}\,dx$ $\qquad \int \frac{u'}{\sqrt{a^2-u^2}}\,dx = \sin^{-1}\left(\frac{u}{a}\right)+C$

$= \sin^{-1}\left(\frac{\sin x}{2}\right)+C$

(25) $I = \int \frac{1}{\sqrt{25-16x^2}}\,dx$

$= \frac{1}{4}\int \frac{4}{\sqrt{5^2-(4x)^2}}\,dx$ $\qquad \int \frac{u'}{\sqrt{a^2-u^2}}\,dx = \sin^{-1}\left(\frac{u}{a}\right)+C$

$= \frac{1}{4}\sin^{-1}\left(\frac{4x}{5}\right)+C$

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(29) $I = \int \frac{1}{x(1+\ln^2 x)}\,dx$

$= \int \frac{1/x}{1^2+(\ln x)^2}\,dx$ $\qquad \int \frac{u'}{a^2+u^2}\,dx = \frac{1}{a}\tan^{-1}\left(\frac{u}{a}\right)+C$

$= \tan^{-1}(\ln x) + C$

(26) $I = \int \frac{1}{x\sqrt{4-9\ln^2 x}}\,dx$

$= \frac{1}{3}\int \frac{3\ 1/x}{\sqrt{2^2-(3\ln x)^2}}\,dx$ $\qquad \int \frac{u'}{\sqrt{a^2-u^2}}\,dx = \sin^{-1}\left(\frac{u}{a}\right)+C$

$= \frac{1}{3}\sin^{-1}\left(\frac{3\ln x}{2}\right) + C$

(*) $I = \int \frac{e^{2x}}{e^{2x}+1}\,dx$

$= \frac{1}{2}\int \frac{2e^{2x}}{e^{2x}+1}\,dx$ $\qquad \int \frac{u'}{u}\,dx = \ln|u| + C$

$= \frac{1}{2}\ln|e^{2x}+1| + C$

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(*) $I = \int \frac{e^x}{1+e^{2x}}\,dx$

$= \int \frac{e^x}{1^2+(e^x)^2}\,dx$ $\qquad \int \frac{u'}{a^2+u^2}\,dx = \frac{1}{a}\tan^{-1}\left(\frac{u}{a}\right)+C$

$= \tan^{-1}(e^x) + C$

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(27) $I = \int \frac{e^{2x}+e^x}{1+e^{2x}}\,dx$

$= \int \frac{e^{2x}}{1+e^{2x}}\,dx + \int \frac{e^x}{1+e^{2x}}\,dx$

$= \frac{1}{2}\int \frac{2e^{2x}}{1+e^{2x}}\,dx + \int \frac{e^x}{1^2+(e^x)^2}\,dx$

$= \frac{1}{2}\ln|1+e^{2x}| + \tan^{-1}(e^x) + C$

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